

MECHANICAL ENGINEERING

ENGINEERING CAD DESIGN & PRODUCT DEVELOPMENT

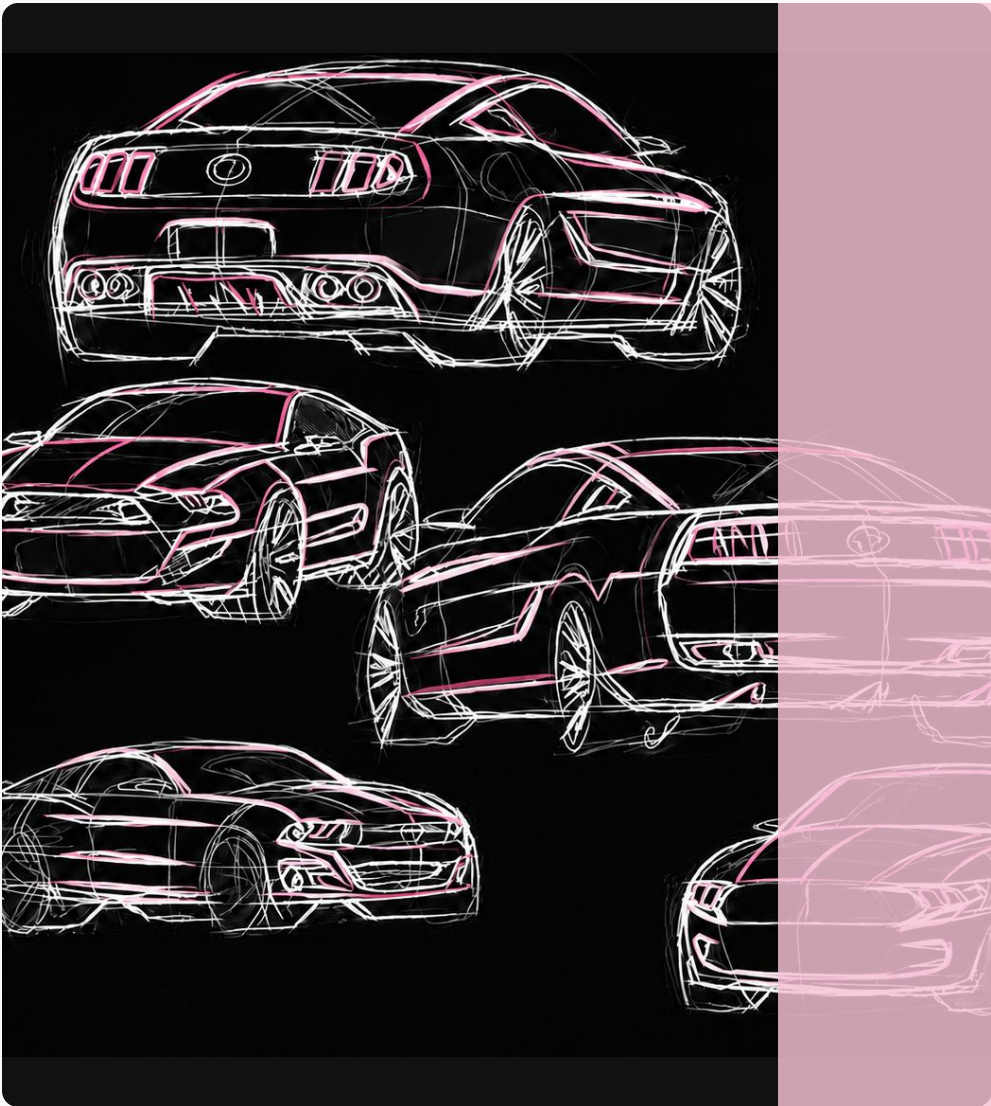
Automotive component development, surface modeling and engineering problem solving, shown through my design work at Flex-N-Gate and engineering projects at Oakland University

EXPLORE MY WORK



SELECTED ENGINEERING WORK / 2025–2026

Automotive CAD portfolio



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01 / PROFILE

ABOUT ME

I am a mechanical engineering student at Oakland University with two Flex-N-Gate internships in Advanced Products CAD Development, tooling, and manufacturing support.

My work starts with a design requirement or a changing vehicle package. I use CATIA to build geometry, review interfaces and communicate the changes needed for a feasible component.

I am especially interested in automotive CAD and product development, where surface intent meets engineering and manufacturing constraints.

[VIEW MY EXPERIENCE](#)



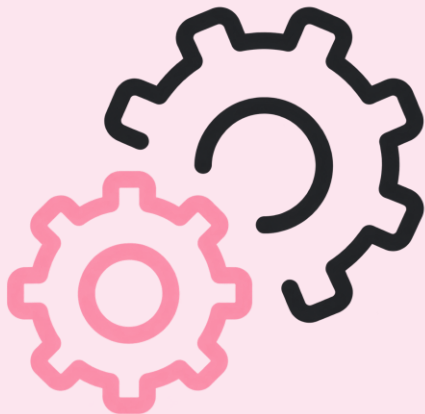
OAKLAND UNIVERSITY

B.S. Mechanical Engineering
Expected May 2028 / GPA 3.63
Rochester, Michigan

02 / NAVIGATION

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03 / AUTOMOTIVE ENGINEERING

EXPERIENCE

FLEX-N-GATE / CAD ENGINEERING INTERN

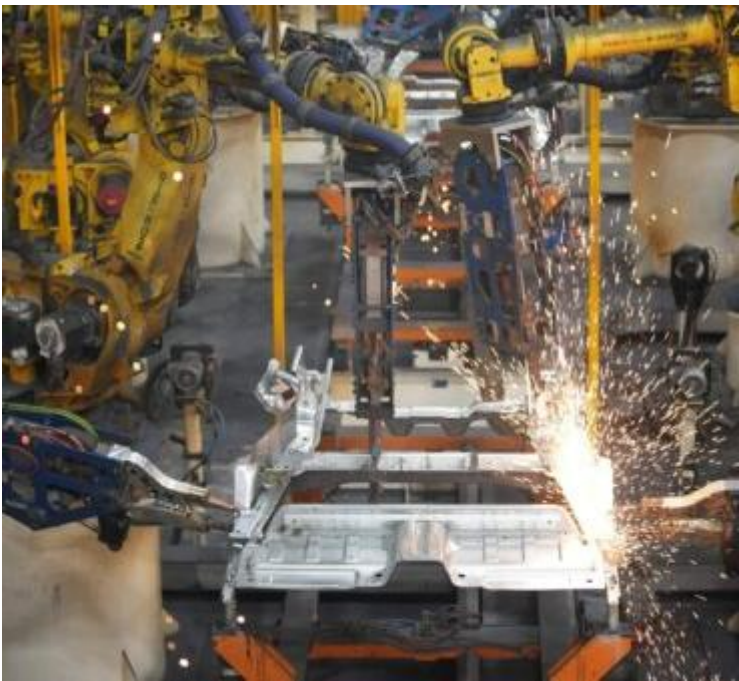
May 2026 - Present / Warren, Michigan

Automotive component development across OEM programs using CATIA V5. Work includes surface and solid modeling, packaging and commonality studies, section development, draft analysis, fixture concepts, and production design revisions.

FLEX-N-GATE / TOOLING ENGINEER INTERN

May - August 2025 / Warren, Michigan

Supported stamped component redesigns and manufacturing problem solving in CATIA. Collaborated across tooling, production, and CAE on component improvements and an integrated headlamp handling design project.



04 / FEATURED PROJECT / 2026

CAMERA BRACKET DEVELOPMENT

THE BRIEF

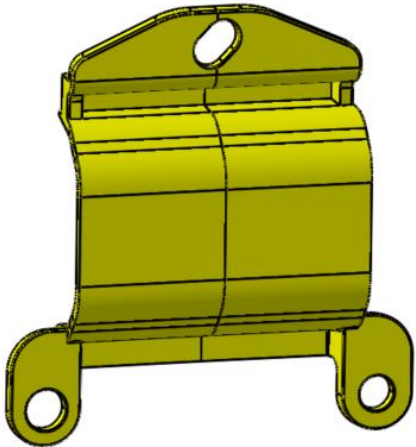
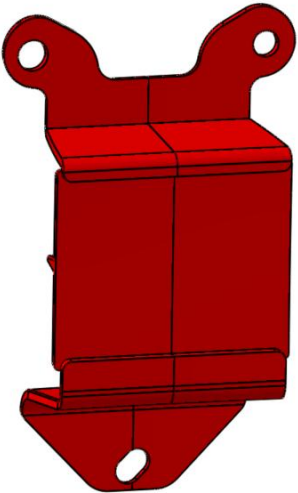
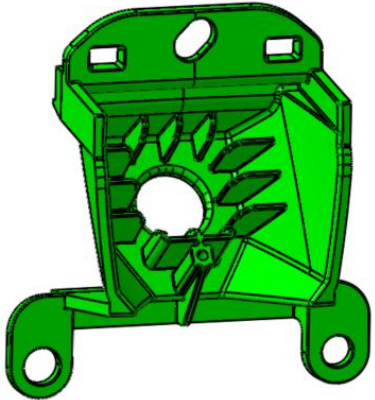
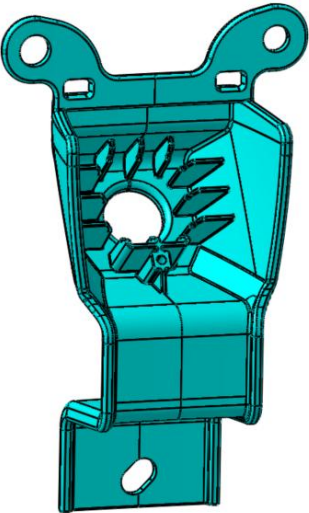
Develop camera bracket concepts for different grille configurations while maintaining camera position and working within the available packaging space

MY CONTRIBUTION

Built the component foundation from supplied studio surfaces, then developed structural ribs, screw bosses and retaining geometry in CATIA.

ENGINEERING CONSIDERATIONS

Stiffness, material use, attachment strategy, assembly access, surrounding package space, and tooling feasibility guided the feature layout.



04 / CAMERA BRACKET / DESIGN DEVELOPMENT

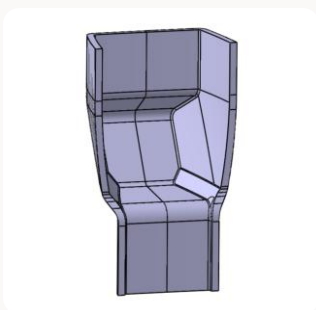
SURFACE TO COMPONENT

I thickened the provided studio surface to 3.5 mm, then developed the functional geometry around the camera package and existing attachment strategy.

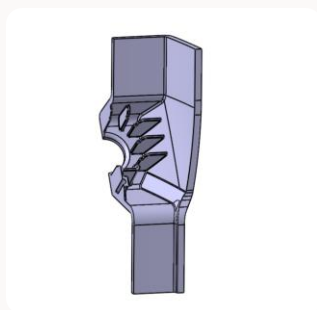
VARIANT 1



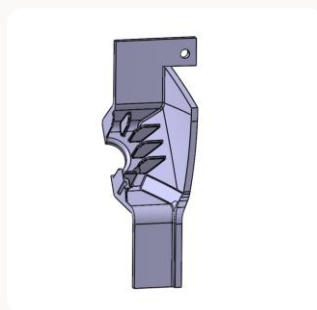
STUDIO SURFACE



3.5 mm FOUNDATION



RIB GEOMETRY



ATTACHMENT DESIGN

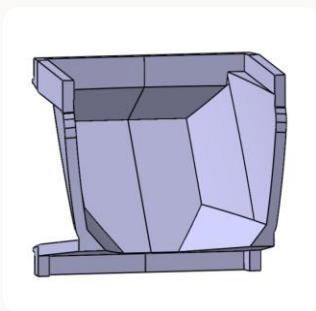


RADII & MIRROR

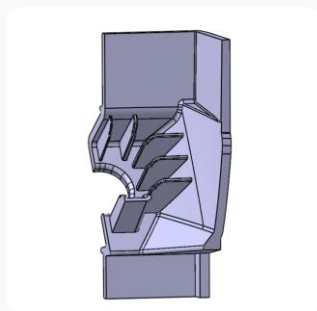
VARIANT 2



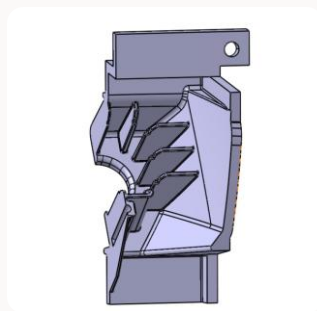
STUDIO SURFACE



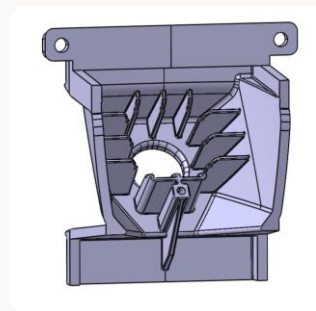
3.5 mm FOUNDATION



RIB GEOMETRY



ATTACHMENT DESIGN

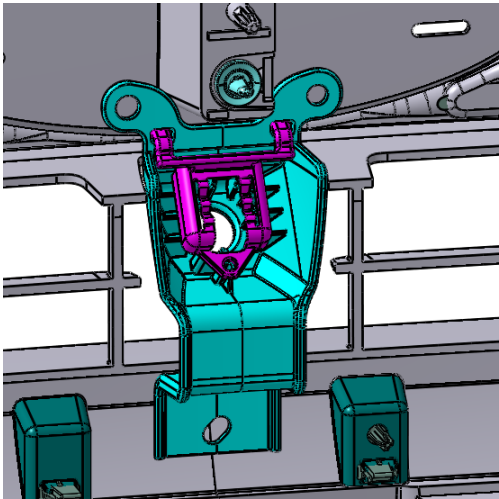


RADII & MIRROR

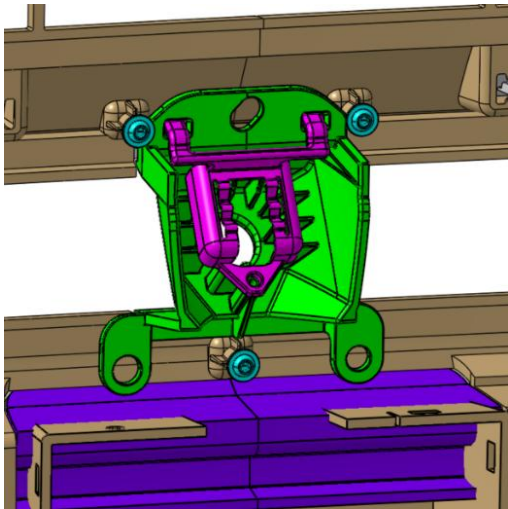
04 / INTERFACES ACROSS VARIANTS

COMMON RETAINER STUDY

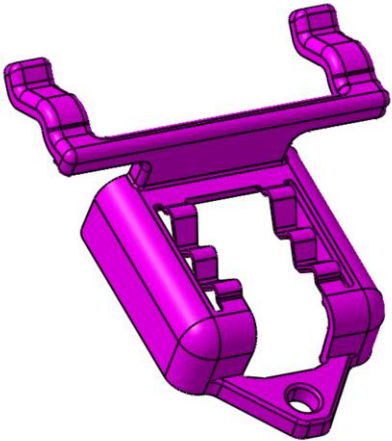
I redesigned retainer and bezel interfaces to investigate a common part across camera and camera-delete configurations, then expanded the study to five variant groups.



VARIANT 1 / PACKAGE



VARIANT 2 / PACKAGE



RETAINER GEOMETRY

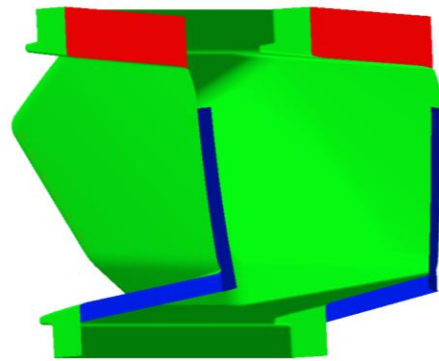
WHAT CHANGED

I compared the available package space across the different configurations and revised the attachment locations, retainer geometry, and surrounding interface features to improve fit and support greater part commonality.

COMMONALITY LIMIT

Three additional bezel configurations used molded-in attachment features, which limited the existing attachment strategy. I began evaluating a mechanical-fastener concept to support commonality across the designs.

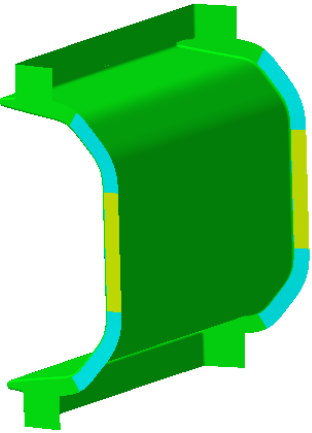
TOOLING FEASIBILITY



CAMERA BEZEL / DRAFT ANALYSIS

MY CONTRIBUTION

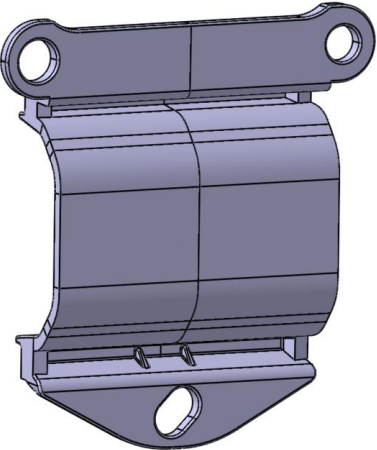
Used draft analysis to find insufficient or negative draft. Revised wall angles and transitions while maintaining material thickness, minimum radii and styling intent.



NON-CAMERA BEZEL / SIDE-FLANGE REVIEW

DESIGN REQUIREMENTS

The design had to meet a 5.4° minimum draft requirement for graining and tooling, along with a 3.5 mm material thickness and 0.5 mm minimum radius.



NON-CAMERA BEZEL / FINAL GEOMETRY

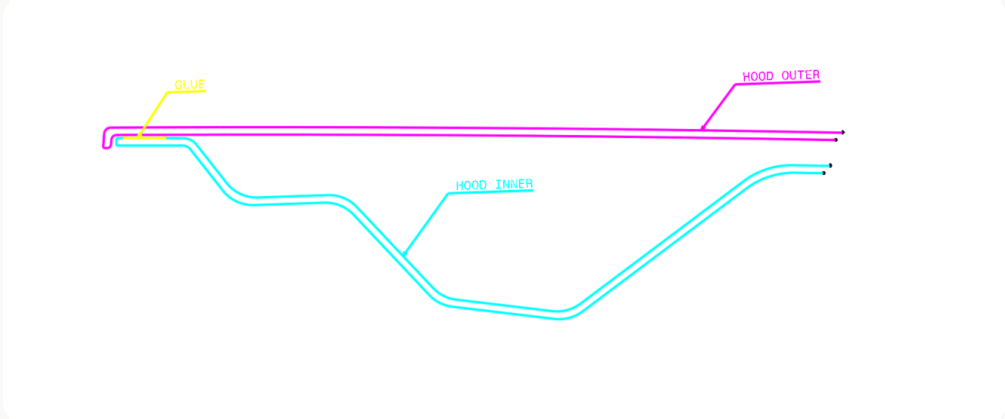
NON-CAMERA VERSION

Used common mounting locations. Narrowed the blanking feature to the plinth pocket and extended the support rib during review.

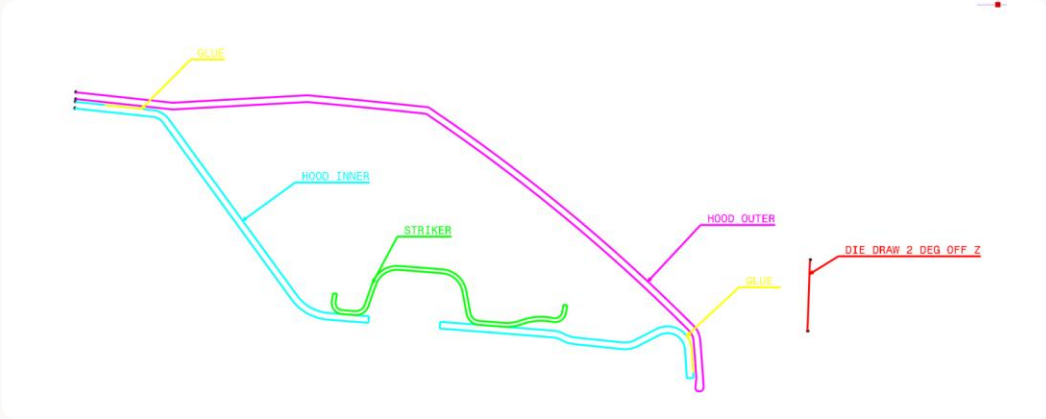
05 / SMC HOOD & FENDER / 2026

STUDIO SURFACES INTO ENGINEERING SECTIONS

I developed five hood and fender sections using early studio surfaces, SIR location data and reference geometry from a surrogate program. The goal was to translate styling surfaces into engineering sections that could be reviewed for packaging, mounting conditions, and manufacturability.



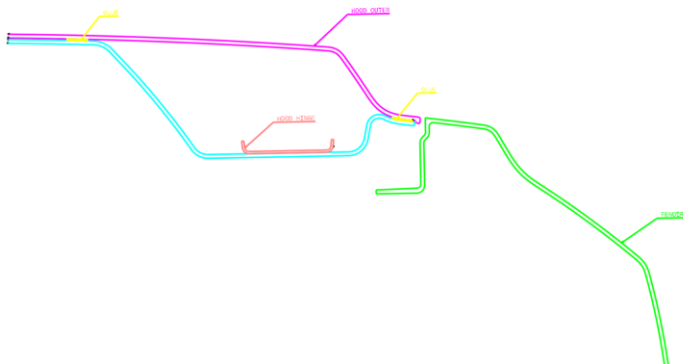
601A / SECTION GEOMETRY



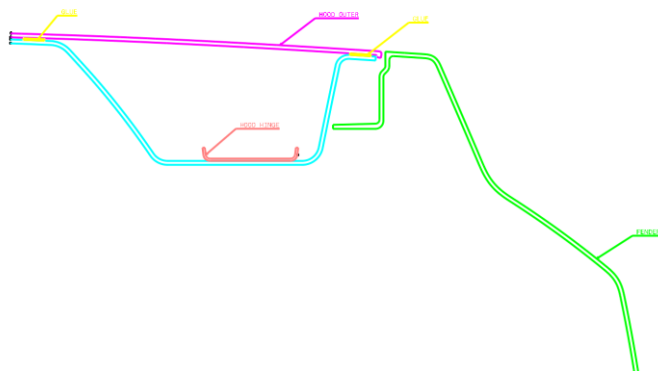
607A / SECTION GEOMETRY

SMC THICKNESS	ADHESIVE WIDTH	MINIMUM RADII	DRAFT REQUIREMENTS	MOUNTING CONDITIONS
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DESIGN DEVELOPMENT THROUGH REVISIONS



605A / INITIAL DEVELOPMENT



605A / UPDATED SECTION

SECTION 605A

Located the section through the wheel centerline. Overlaid studio surfaces with SIR data, worked with a 3 mm fender-flare thickness assumption and created flats for future doghouse development.

PACKAGE REVIEW

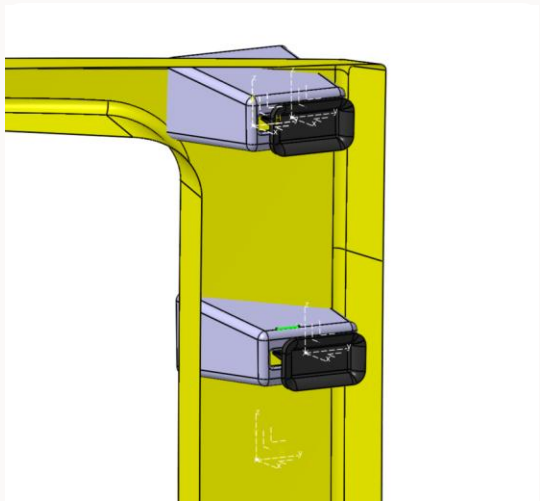
Reviewed the surrounding package using an additional 6 mm clearance target to help avoid tight conditions and maintain usable space around the section.

SURFACE CHANGES

When revised studio data arrived, updated reinforcements, mounting plates and glue locations. Section 703A needed the most rework to remain within the new boundaries.

COMMON HEADLAMP ATTACHMENTS

I evaluated a third and potentially fourth common attachment across grille variants. The available X-direction movement, surrounding headlamp geometry, and attachment clearances limited the feasible design space..



17 mm / INITIAL PROPOSAL



9 mm / REDUCED-WIDTH STUDY

14.5 mm

Approximate minimum additional surround width identified through the packaging study.

DECISION

Studio rejected the 17 mm proposal and proposed 9 mm. The 9 mm option did not provide the required geometry and clearances.

MY CONTRIBUTION

Revised attachment towers and slots, maintained 4 mm X-direction clearance on both sides of the attachment geometry, and developed comparative CAD concepts to communicate the feasible packaging range for review.

06 / ATTACHMENT BRACKET DEVELOPMENT / 2026

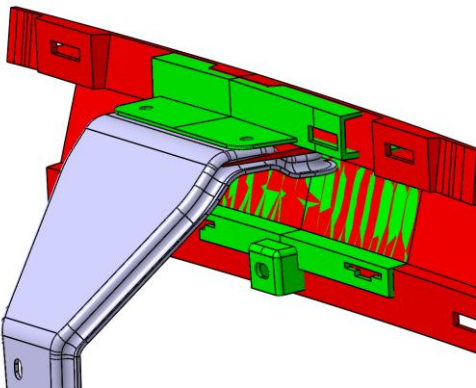
BRACKET DEVELOPMENT FOR A CHANGING FASCIA

MY CONTRIBUTION

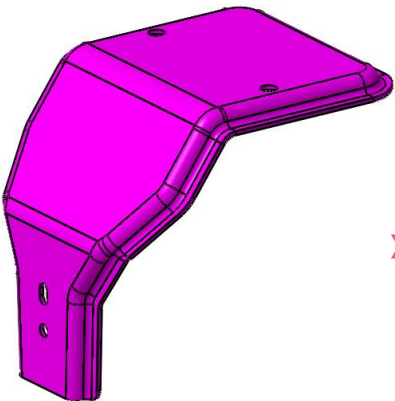
Developed a new attachment bracket after the mounting location shifted by approximately 85 mm. Built the surface geometry in CATIA GSD, added fillets and a locating lip, then thickened the component to 3 mm and developed the final attachment features.

UPDATED PACKAGE CONDITIONS

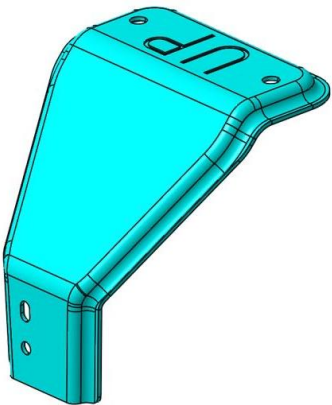
When updated fascia geometry introduced interference, I revised the rear wall, overall height, cutouts, and mounting-hole locations. Repeated package and interface checks helped align the bracket with the revised surrounding geometry.



BRACKET / SURROUNDING INTERFACE

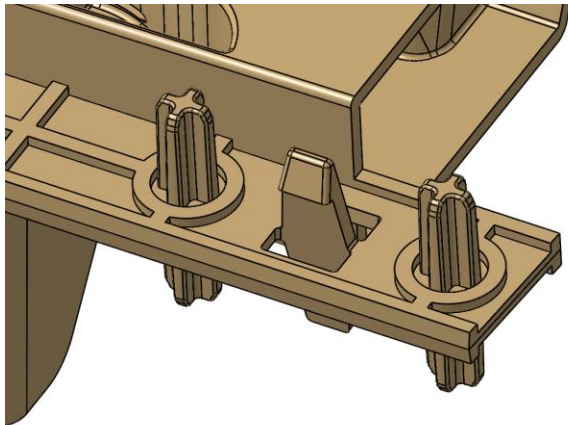


INITIAL BRACKET DESIGN

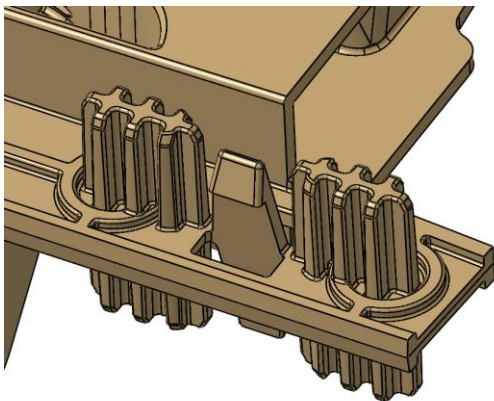


LATEST BRACKET REVISION

MANUFACTURING-DRIVEN DESIGN UPDATES



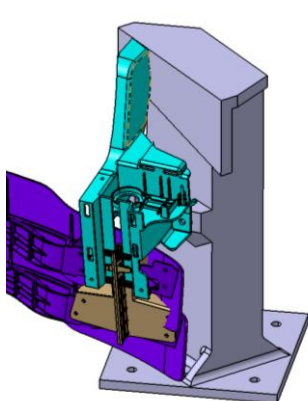
ORIGINAL GUIDE



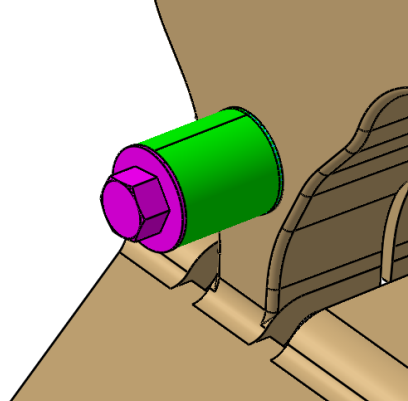
REVISED GUIDE / THREE RIBS

GUIDE-RIB REDESIGN

Extended the center rib and added two cross ribs to strengthen the guide geometry. After identifying interference with the block cover, I refined the design to restore clearance while maintaining the updated rib layout. Applied 1° draft and 0.5–1.0 mm edge fillets.



FIXTURE BLOCK

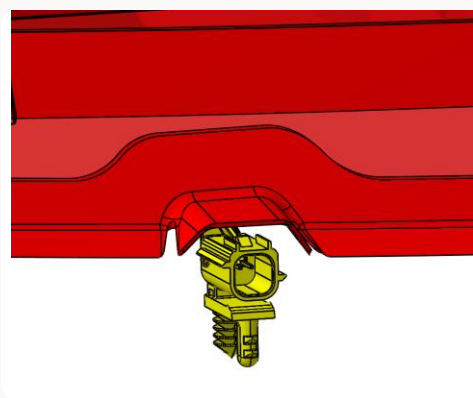


BOLT ADAPTER

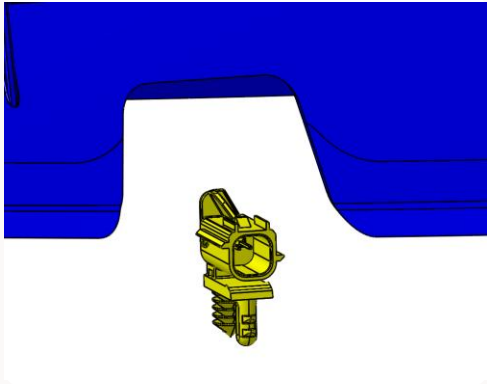
FIXTURE & FASTENER WORK

Designed fixture blocks from reference geometry and engineering requirements, matching the required base dimensions and hole spacing. Also developed a bushing that allowed the existing bolt to be retained.

CLEARANCE & PRODUCTION SUPPORT



ORIGINAL / SENSOR REGION

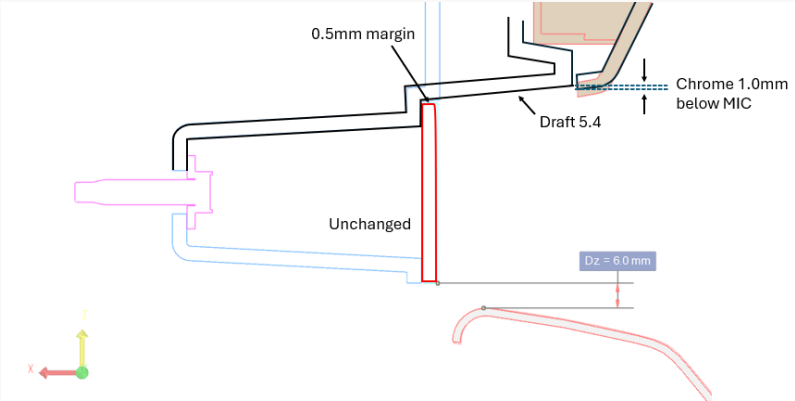


NEW CLEARANCE CUTOUT

SEAL REVISIONS & REWORK SUPPORT

Created a sensor-clearance cutout and production dimensions for a rework fixture. Referenced original tab notches, ribs and molded features to locate the new cutout and its depth.

Additional revisions included a drain opening, shortened seal geometry, shifted clip locations, and updated bolt cutouts.



BOLT-COVER SECTION

MOUNTING-BOLT COVER STUDY

Evaluated the available package space while keeping the bumper location fixed and allowing the bolt/cover assembly to move only in Z. The study had a 26.1 mm socket clearance requirement and 6 mm bumper clearance.

The section was used to determine whether the surrounding trim needed to move and how much usable package space remained.

07 / CAD + CAE PROBLEM SOLVING / 2025

A DEDICATED HEADLAMP GRIP

THE PROBLEM

During assembly, the headlamp attachment tab was being used as a lifting point. The tab needed to remain thin and flexible for installation, but the additional handling load created stress beyond what the feature was intended to support.

DESIGN DIRECTION

I evaluated several approaches, including reinforcement, material changes, and a dedicated handle. I decided on implementing a separate grip feature that redirected the operator away from the flexible attachment tab while providing a more appropriate lifting location.

My role: CATIA geometry development and design iterations with CAE feedback from intern Frank Wieckowski.

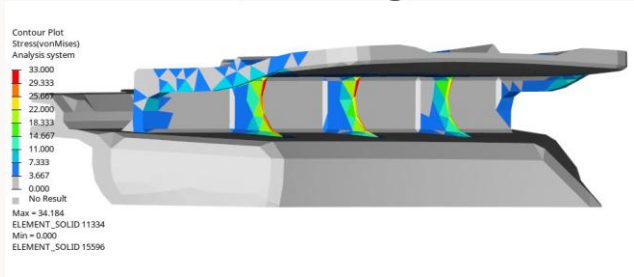


HANDLE / OPERATOR CONTEXT

07 / HEADLAMP HANDLE / DESIGN DEVELOPMENT

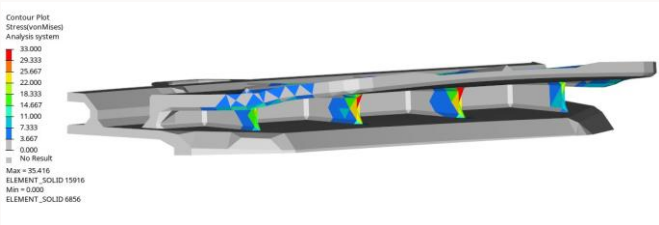
DESIGN ITERATION THROUGH CAE FEEDBACK

I revised rib geometry, transition radii, and the handle back wall as successive CAE results revealed local stress concentrations. Package-driven changes also required additional design updates and renewed simulation review.



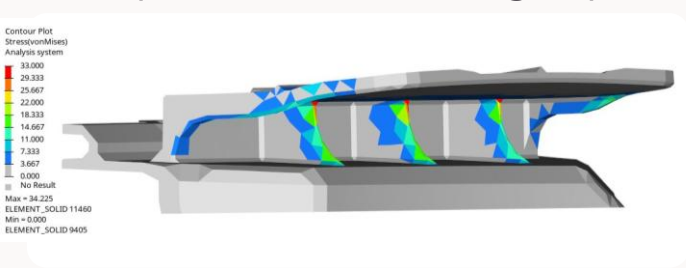
01 / INITIAL HANDLE

Developed the initial handle with cross ribs, corner supports, and long connecting ribs



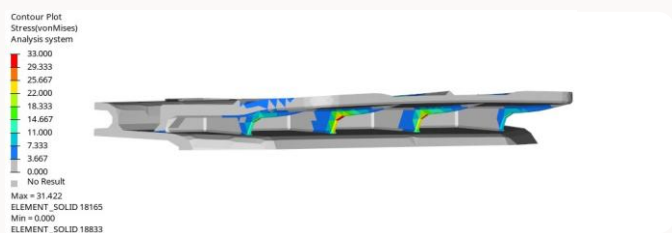
04 / PACKAGE REVISION

Revised the geometry to fit updated package. Conditions however, the change increased local stress and required further refinement.



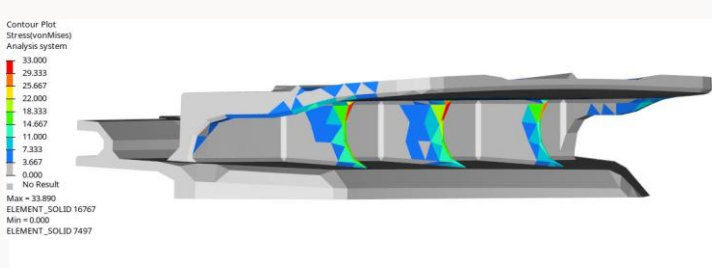
02 / LOCAL TRANSITIONS

Adjusted rib offsets, heights, and radii to reduce localized stress concentrations.



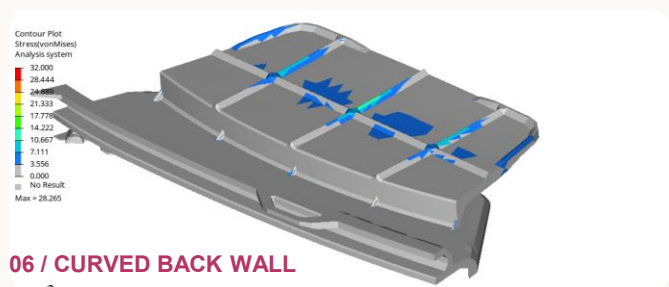
05 / RIB EXTENSION

Extended the rib to the handle perimeter and added a tangent top transition to improve the load path.



03 / CURVED CONNECTION

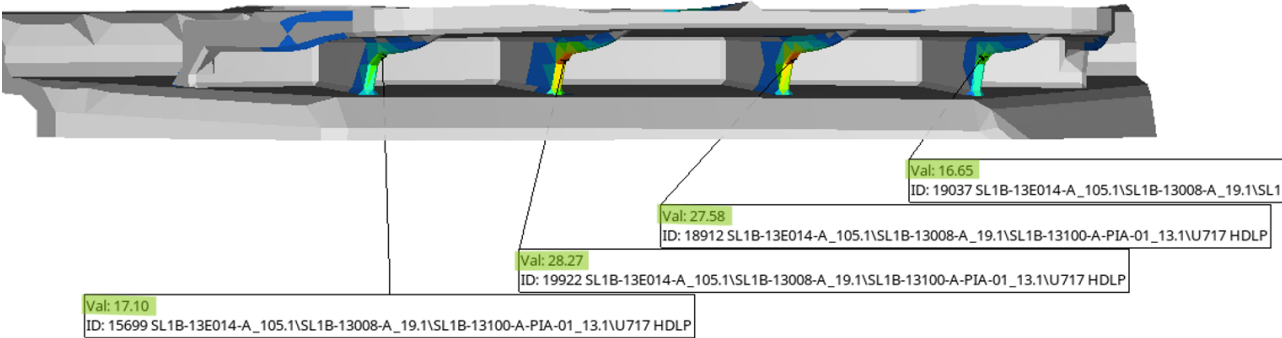
Replaced a perpendicular rib connection with a curved transition to improve load distribution.



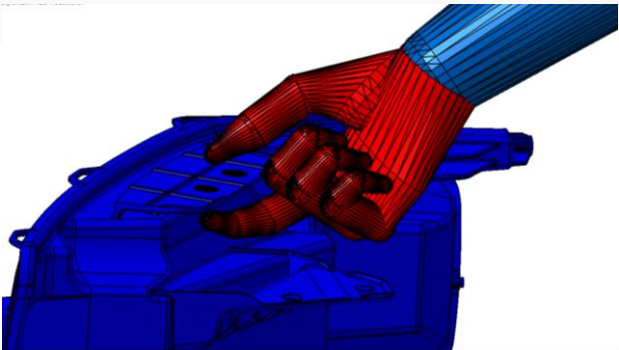
06 / CURVED BACK WALL

Revised the rear wall with a curved profile to reduce the remaining stress around the support ribs.

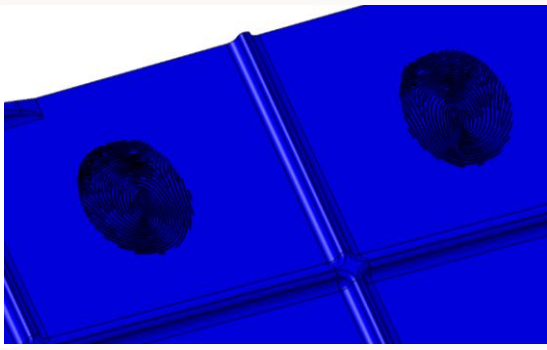
FINAL DESIGN & SIMULATION



FINAL HANDLE / STRESS CONTOUR



FINAL CAD / HANDLE GEOMETRY



FINGER GROOVE DETAIL

Selected internship work. Flex-N-Gate and respective customer IP.

CAE results shown represent simulation-based validation of the final concept.

~28 MPa

Final peak stress under the defined simulation load.

STUDY CRITERION

The final design remained below the study's 33 MPa material-yield threshold. The highest stress occurred at the two center support ribs, while the rest of the model remained below approximately 17.1 MPa.

MODEL CONDITIONS

70 N upward lift load applied across a 90 mm grasp region, with gravity and fixed boundary constraints used to represent the lifting condition.

FINAL DESIGN FEATURES

The handle was positioned 45 mm above the attachment tab for operator access. A fingerprint groove pattern helped guide hand placement, while 1.5 × 1.5 mm cross ribs and corner supports were used to reduce handle displacement and improve stiffness.

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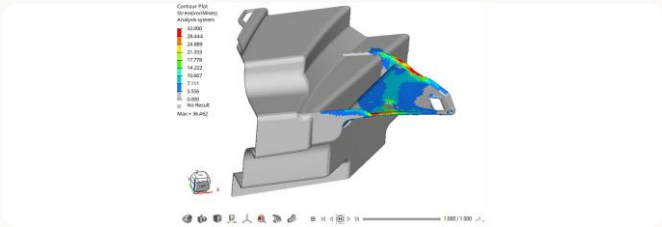
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07 / HEADLAMP HANDLE / SIMULATION ANIMATIONS

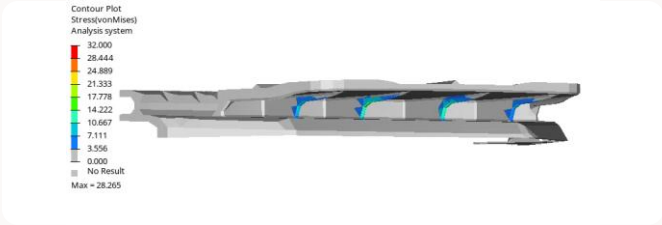
CAE SIMULATION LIBRARY

These animations show the CAE results used throughout the design process, from the original attachment tab stress condition through the final handle design and operator simulation. Select a clip to view the full animation in the digital portfolio.



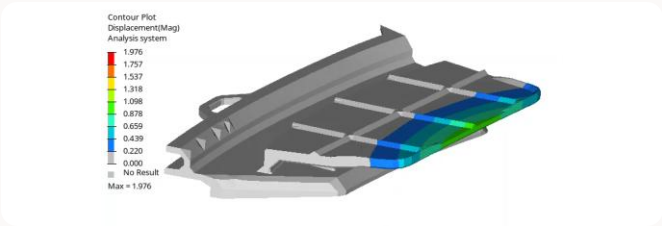
ORIGINAL TAB / STRESS CONDITION

PLAY CLIP >



FINAL HANDLE / STRESS CONTOUR

PLAY CLIP >



FINAL HANDLE / DISPLACEMENT

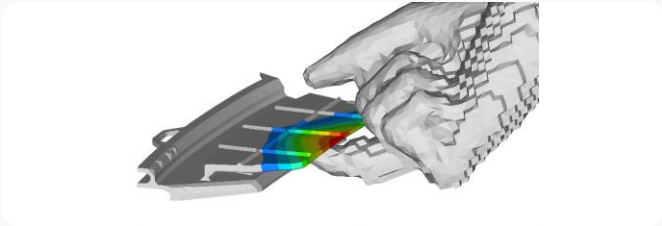
PLAY CLIP >



OPERATOR CONTEXT / STRESS CONTOUR

PLAY CLIP >

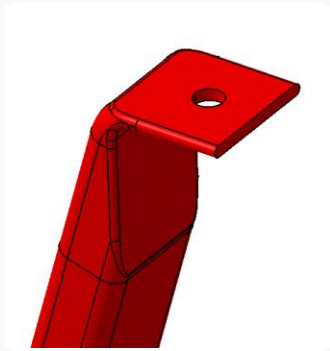
*Contour visualization only
(stress values not used for valuation)



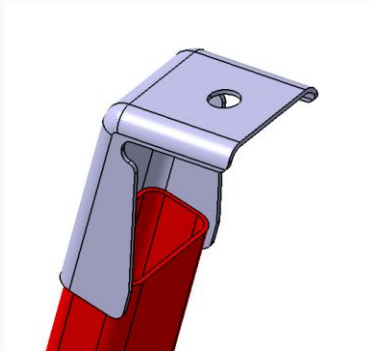
OPERATOR CONTEXT / DISPLACEMENT

PLAY CLIP >

DESIGN FOR MANUFACTURABILITY



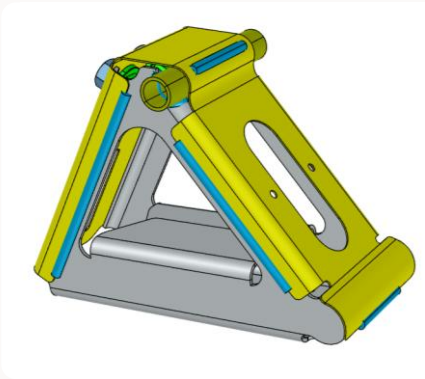
ORIGINAL END



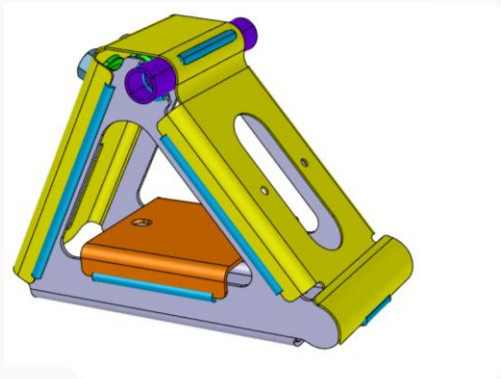
NEW BRACKET

FUEL TANK STRAP REDESIGN

Shortened and separated the original tube ends, then developed a stamped bracket using the existing dimensional requirements. The revised geometry created weld access on three sides while maintaining the required attachment condition.



ORIGINAL RAIL TURNER



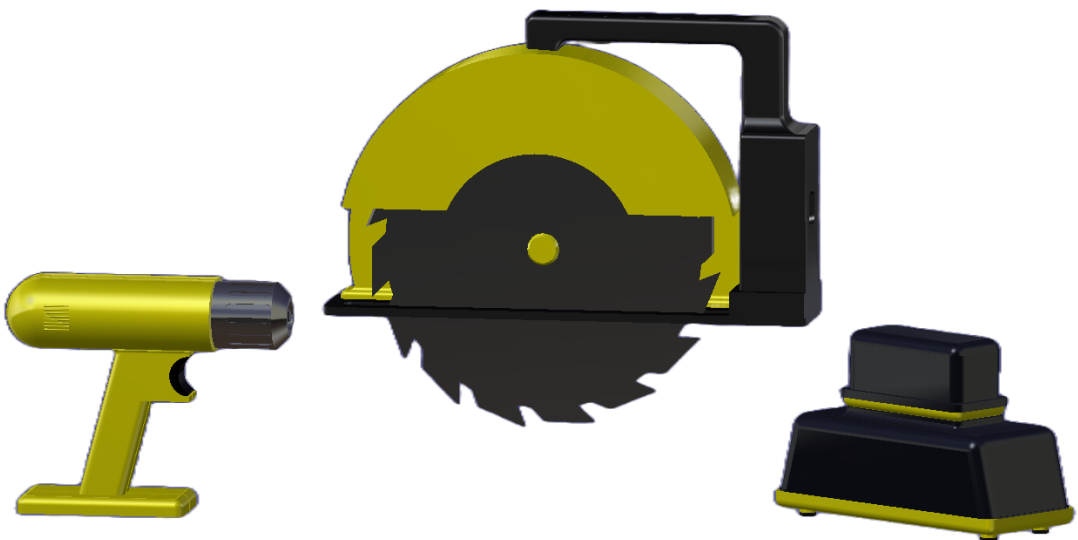
PROPOSED RAIL TURNER

RAIL TURNER REDESIGN

Removed the folded edge features and redesigned the lower bracket as a separate diagonal component to provide more accessible side welds. The concept was reviewed by production and approved with minor revisions.

08 / EGR 1200 / OAKLAND UNIVERSITY

ACADEMIC CAD: POWER-TOOL KIT



SAW / DRILL / BATTERY CHARGER

The project included dimensioned sketches, individual part models, constrained subassemblies, and a final assembly combining the saw, drill, and battery charger.

Academic team project

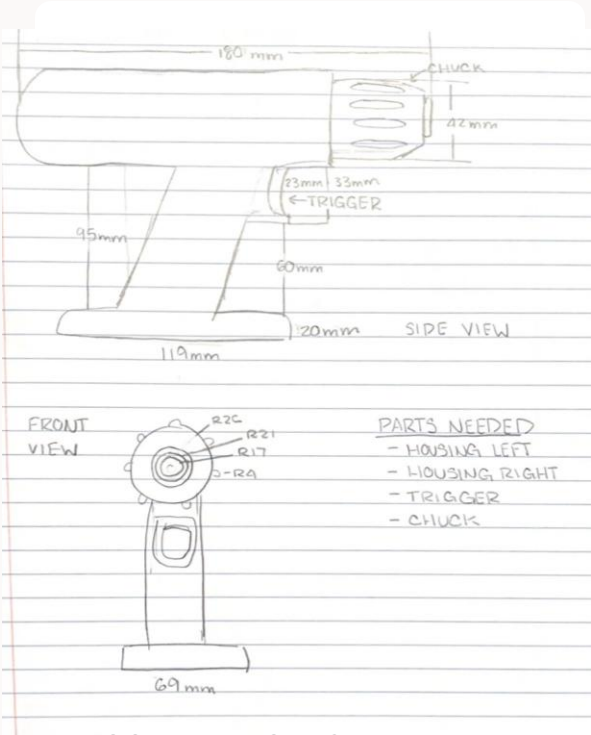
MY CONTRIBUTION

Contributed to the CAD modeling and assembly development of the team's multi-component power-tool kit. Used parametric modeling, dimensional control, and assembly constraints to develop individual components and combine them into functional subassemblies.

TEAM

Cadenn Clower
Emma Trajkovski
Fernanda Ferraz

DIMENSIONS TO ASSEMBLY



DIMENSIONED DRILL SKETCH

My contribution focused researching and drafting the dimensioned sketches into CAD geometry and developing the constrained assembly for the cordless drill.



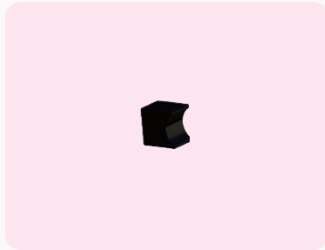
LEFT HOUSING



CHUCK



RIGHT HOUSING



TRIGGER



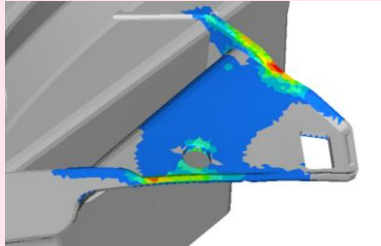
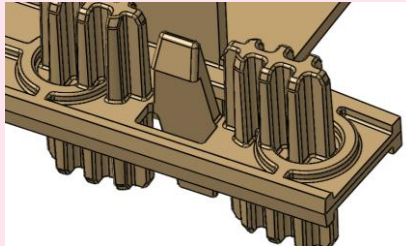
CONSTRAINED ASSEMBLY

ASSEMBLY LOGIC

The assembly was built by fixing the left housing as the base component, then aligning the chuck using coincidence and contact constraints. The trigger was positioned within its molded slot, and the right housing was constrained to complete the assembly. An exploded view was used to verify component relationships and constraint placement.

SKILLS

Technical tools, engineering methods, and design skills developed through my internships, CAD projects, and mechanical engineering coursework.



CAD & DESIGN

CATIA V5 / 3DEXPERIENCE
Generative Shape Design (GSD)
Part Design
Surface + Solid Modeling
Assembly Development
Class A Surfacing
SolidWorks / AutoCAD

MANUFACTURING

Design for Manufacturability
Draft Analysis
GD&T
Packaging + Interference Studies
Fixture Design
Tolerance Assessment

ANALYSIS + ENGINEERING

ANSYS / FEA
Structural analysis
Design verification
MATLAB / Spinfire
Thermodynamics
Heat transfer / Fluid mechanics

ENGINEERING PRACTICE

Technical communication
Design Reviews + Presentations
Root-cause analysis
Cross-Functional Collaboration
Microsoft Office
C++ / Verilog

10 / ENGINEERING BEYOND THE DESK

EDUCATION & INVOLVEMENT

OAKLAND UNIVERSITY

- GPA: 3.63
- Dean's List: All Semesters

LEADERSHIP + RECOGNITION

- Youth Volunteer Leader / Stoney Church
- Academic All-American / 2024

At A. L. Holmes Academy, I helped students adjust and test water-rocket fin designs between launches, giving me another chance to apply hands-on problem solving and design iteration.

At Focus HOPE, I volunteered with other interns and employees during a food-packaging event, strengthening teamwork, communication, and organization in a fast-paced group setting.



A. L. HOLMES / WATER ROCKETS



FOCUS HOPE / FOOD PACKAGING



GET IN TOUCH

Mechanical engineering student with automotive CAD and product development experience, focused on developing components around real design, packaging, and manufacturing requirements.

cadennclower@gmail.com

586-405-0598

Macomb, Michigan
Oakland University
B.S. Mechanical Engineering / May 2028

[BACK TO MY WORK](#)[VIEW RESUME](#)[LINKEDIN](#)

Cadenn Clower

Macomb, MI | 586-405-0598 | cadennclower@gmail.com

Professional Summary:

Mechanical engineering student with two Flex-N-Gate internships in design, structural analysis, and manufacturing support. Seeking internships in design, analysis, or manufacturing. Experience includes CATIA V5, 3DEXPERIENCE, ANSYS, GD&T, packaging studies, prototype testing, fixture development, and design reviews.

Education:

Bachelor of Science in Mechanical Engineering | Expected May 2028

Oakland University | Rochester, MI | GPA: 3.63

Engineering Experience:

CAD Engineering Intern

May 2026 - Present

Flex-N-Gate | Warren, MI

- Designed a Nissan Pathfinder bracket in CATIA V5 GSD and Part Design for an approximately 85 mm mounting shift; added locating/attachment features and resolved interference with revised customer fascia.
- Evaluated F-250 headlamp packaging alternatives; identified an approximately 14.5 mm minimum width increase while maintaining 4 mm attachment clearances, and prepared comparative CAD sections for program review.
- Developed a common camera retainer for five F-250 variant groups, now in final preproduction; evaluated camera and camera-delete configurations, fastener concepts, and bezel interfaces to improve part commonality.
- Developed Class A surfaces for multiple camera-bezel designs in CATIA V5; performed draft analysis on three configurations, evaluating wall thickness and radii while checking mold-release feasibility.
- Designed fixture blocks and a bolt-adaptor bushing to reuse an existing fastener; defined seal-rework fixture dimensions, referencing original notches, ribs, and molded features for cutout position and depth.
- Redesigned an active-air-dam guide with three ribs and resolved interference between the guide and its cover.
- Developed five hood/fender cross sections for sheet molding compound (SMC) with thickness, adhesive-width, radius, and draft requirements; revised reinforcement and mounts as customer studio surfaces changed.

Tooling Engineer Intern

May 2025 - Aug 2025

Flex-N-Gate | Warren, MI

- Designed and optimized an F-150 headlamp attachment using ANSYS FEA; reduced calculated peak stress from 34 to 17.1 MPa, a 16.9 MPa reduction (approximately 50%), and confirmed stress below material yield limits.
- Designed and revised stamped-steel components in CATIA V5 for Ford, GM, and Mercedes programs; applied DFM while evaluating material properties, dimensions, tooling constraints, and production feasibility.
- Collaborated with engineering, tooling, and production teams to investigate manufacturing issues, perform root-cause analysis, and develop corrective actions supporting quality, reliability, safety, and process efficiency.

Engineering Projects:

CAD Drill Kit Design Project: Designed and assembled a multi-component drill kit in CATIA V5 using 3D parametric modeling, material selection, dimensional control, assembly constraints, and engineering drawing standards.

Additional Projects: Regenerative Steam Power Plant Analysis; R-134a Thermodynamic System Analysis; Manufacturing Process Simulation & Optimization; Arduino General-Purpose Input/Output (GPIO), Pulse-Width Modulation (PWM) LED Control & Binary Counter Programming.

Technical Skills:

CAD & Analysis: CATIA V5, 3DEXPERIENCE, SolidWorks, AutoCAD, ANSYS, MATLAB, Spinfire

Design & Manufacturing: Class A surfacing, Generative Shape Design (GSD), Part Design, surface/solid modeling, assembly development, geometric dimensioning and tolerancing (GD&T), design for manufacturability (DFM), draft analysis, fixture design

Engineering Methods: Finite element analysis (FEA), structural analysis, design verification, prototype testing, tolerance assessment, root-cause analysis, thermodynamics, fluid mechanics, heat transfer

Programming & Productivity: C++, Verilog, Microsoft Office

Activities & Achievements:

Youth Volunteer Leader - Stoney Church | Academic All-American (2024) | Dean's List - All Semesters